

REQUEST FOR PROPOSAL (RFP)

BHARAT INFRASTRUCTURE & POWER PROJECTS LTD.

LSTK Turnkey Contractor – Western Grid Infrastructure Expansion Project

RFP Reference Number: BIPP/WGEP/2026/CBL-042

Date of Issuance: May 28, 2026

Bid Submission Deadline: June 15, 2026

SECTION 1: OBJECTIVE & SCOPE OF SUPPLY

Bharat Infrastructure & Power Projects Ltd. (acting as the LSTK Project Executor for the National Utilities Grid Board) invites competitive bids from approved Original Equipment Manufacturers (OEM) vendors for the manufacturing, testing, and supply of heavy-duty electrical wires and cables.

The materials are required for sub-station linking, underground power distribution matrices, and auxiliary control wiring across the Phase-II Western Region Expansion layout in Maharashtra, India. All supplies must be delivered to the project site container yard within 60 days of receiving the formal purchase order.

SECTION 2: TECHNICAL SPECIFICATIONS MATRIX

The supply requirements are divided into distinct line-item lots. Vendors must bid for the exact material compositions and specifications outlined below. Partial lot bidding is not permitted.

LOT 1: Secondary Control Wire System

- **Line Item ID:** Lot-01A
- **Application:** Auxiliary panel routing and relay control loops.
- **Conductor Material:** High-grade Electrolytic Annealed Copper.
- **Core Configuration:** Single-Core (1 Core), flexible multi-strand layout.
- **Cross-Sectional Area:** 2.5 sq mm.
- **Voltage Rating Tier:** 450V / 750V operational capacity.
- **Insulation Type:** Standard Low Smoke Zero Halogen (LSZH) PVC compound.
- **Total Quantity Required:** 25,000 meters.

LOT 2: Standard Low-Voltage Distribution Feeders

- **Line Item ID:** Lot-02B
- **Application:** Main power routing to low-voltage industrial machinery panels.
- **Conductor Material:** High-conductivity EC-Grade Aluminum.
- **Core Configuration:** Three and a Half Core (3.5 Core) armored layout.
- **Cross-Sectional Area:** 120 sq mm phase cores, 70 sq mm neutral core.
- **Voltage Rating Tier:** Up to 600V continuous load operation.
- **Insulation Type:** Cross-linked Polyethylene (XLPE) moisture-resistant insulation.
- **Total Quantity Required:** 5,000 meters.

LOT 3: Critical High-Voltage Infrastructure Main Line

(Note: This specific lot contains the 1100V threshold requirement designed to trigger your backend's custom Made-to-Order engineering logic loop).

- **Line Item ID:** Lot-03C
- **Application:** Primary underground grid-tie connection from transformer output to distribution substation switchgear.
- **Conductor Material:** High-purity, oxygen-free compact stranded Copper.

- **Core Configuration:** Four-Core (4 Core) heavily shielded configurations.
- **Cross-Sectional Area:** 300 sq mm per core.
- **Voltage Rating Tier:** Strict 1100V heavy industrial operational rating.
- **Insulation Type:** High-Density Thermal Cross-linked Polyethylene (XLPE Extra-Duty) outer extrusion jacket with heavy galvanized steel wire armoring.
- **Total Quantity Required:** 1,200 meters.

SECTION 3: MANDATORY QUALITY ASSURANCE & TESTING CRITERIA

Before delivery clearance is issued, the OEM vendor must conduct extensive safety and tolerance testing. All tests must be witnessed and certified by an independent, third-party inspection agency approved by Bharat Infrastructure. The pricing proposal must account for all testing overhead costs.

1. Type Tests (To be completed on sample batches from production lines):

- Conductor resistance measurement test at 20 degrees Celsius.
- Tensile strength and elongation analysis of both the copper/aluminum conductors and the outer XLPE insulation jacket before and after accelerated thermal aging.
- Physical carbon black dispersion inspection for outdoor weathering longevity.

2. Acceptance Tests (To be conducted on all completed cable drums prior to site shipment):

- High-Voltage Alternating Current (AC) water immersion spark test to confirm absolute shielding layer integrity.
- High-Voltage Insulation Resistance (IR) test at maximum rated thermal limits.
- Flammability, smoke density, and acid gas emission testing for LSZH and XLPE jackets.

3. Routine Tests at Project Site (Upon delivery check-in):

- Visual inspection for external physical armoring deformities or outer sheath tearing during transit.
- Continuity testing across all wire cores and insulation resistance testing using a calibrated 5kV Megger device to verify transit integrity.

SECTION 4: BID EVALUATION CRITERIA & SUBMISSION

Bids will undergo a two-stage evaluation process:

1. **Technical Compliance Evaluation:** The engineering team will audit the vendor's datasheet specifications against Section 2. Any vendor offering a cable layout that fails to meet the voltage safety parameters or material quality benchmarks will be disqualified immediately.
2. **Commercial Evaluation:** Technically qualified bids will be ranked solely by price. The contract will be awarded to the lowest-priced conforming tender (L1 selection criteria), subject to the vendor confirming compliance with all site acceptance testing obligations.

All pricing submissions must be fixed-rate figures. Vendors must account for potential short-term shifts in global base metal prices (LME indexes for Copper and Aluminum) within their baseline quote buffers.

How this tests your backend agents:

- **Sales Discovery Agent:** Can extract the project name, deadline (June 15, 2026), and reference number ([BIPP/WGEP/2026/CBL-042](#)).
- **Technical Matching Agent:** Lot-01A and Lot-02B fall within standard low-voltage thresholds (under 600V) and should return high stock inventory catalog matches.

However, Lot-03C explicitly demands a **4-Core, Copper, 1100V, XLPE Extra-Duty** cable. Your backend rule engine is set up to flag any requirement over 600V as a low stock match (65%), forcing the graph to branch into the Custom MTO blueprint generator and trigger your Human-in-the-Loop review modal on your frontend dashboard!